Machining Process

- Manufacturing processes can be broadly divided into two groups:

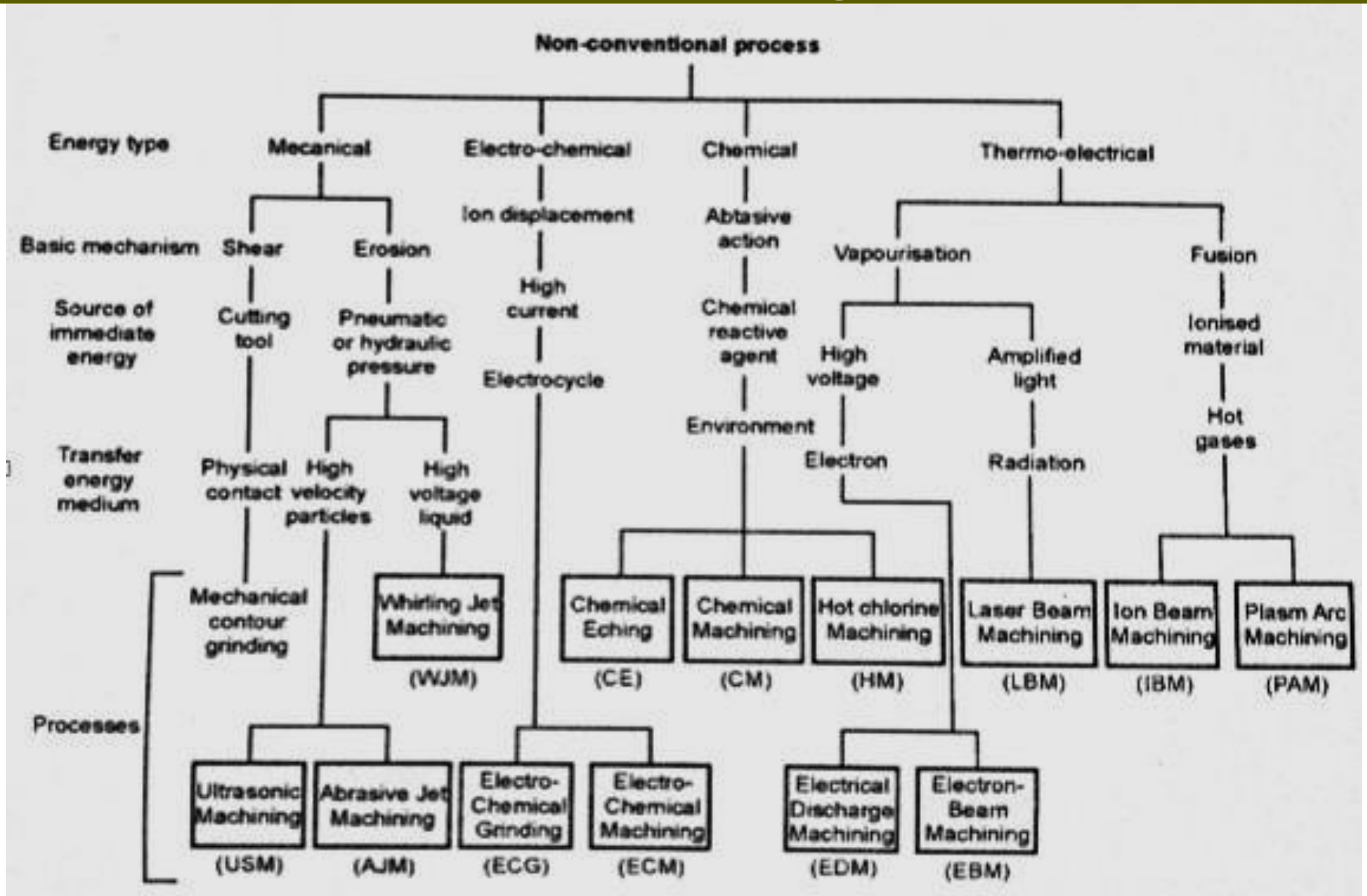
a) Primary manufacturing processes : Provide basic shape and size

b) Secondary manufacturing processes : Provide final shape and size with tighter control on dimension, surface characteristics

- Material removal processes once again can be divided into two groups

1. Conventional Machining Processes
2. Non-Traditional Manufacturing Processes or Non-conventional Manufacturing processes

Classification of Non-conventional Machining Process



Needs for Non-Traditional Machining

- Extremely hard and brittle materials or difficult to machine materials are difficult to machine by traditional machining processes.
- When the workpiece is too flexible or slender to support the cutting or grinding forces.
- When the shape of the part is too complex.
- Intricate shaped blind hole – e.g. square hole of 15 mm x 15 mm with a depth of 30 mm
- Machining of composites.

Conventional Machining Vs Non-conventional Machining Process

- In Conventional machining process, the cutting tool and workpiece are always in physical contact, with a relative motion against each other, which results in friction and a significant tool wear.
- In Non-conventional machining processes, there is no physical contact between the tool and workpiece. Although in some non-traditional processes tool wear exists, it rarely is a significant problem.

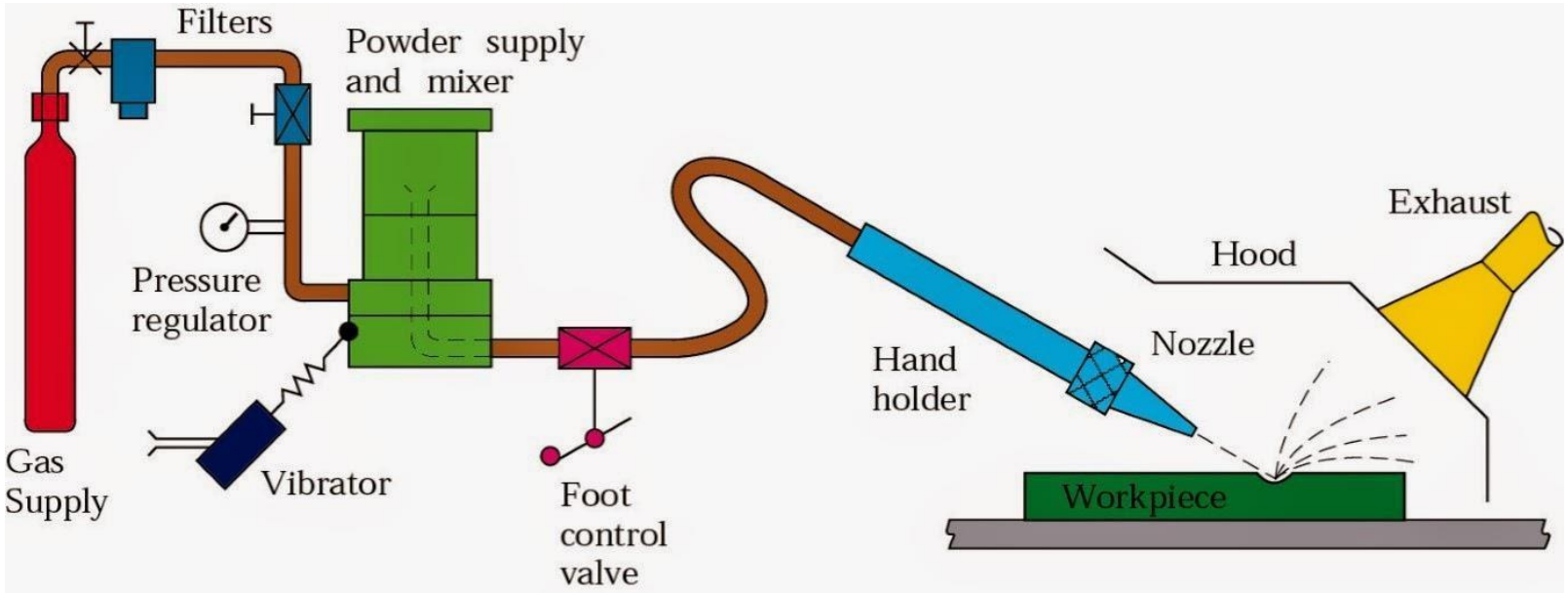
Abrasive Jet Machining (AJM)

- In AJM, the material removal takes place due to impingement of the fine abrasive particles. These particles move with the high speed air (or gas) stream.
- The abrasive particles are typically of 0.025 mm diameter and the air discharges at a pressure of several atmosphere.

Abrasive Jet Machining (AJM)

Abrasives:- Aluminum oxide, Boron Nitride, Diamond Dust

Dry air, Nitrogen or CO_2



Working of Abrasive jet machining

- The nozzle is made of a hard material like Tungsten Carbide here fine grained abrasive particles are fed from the Hooper into the mixing chamber.
- High pressure air is forced in to the mixing chamber.
- The stream of abrasive particles bombards the workpiece at a very high speed and removes the work material due to erosion.
- The abrasive particle feed rate is controlled by the amplitude of vibration of the mixing chamber.

Typical AJM Parameters

- Abrasive
  - Aluminum oxide for Al and Brass.
  - SiC for Stainless steel and Ceramic
  - Bicarbonate of soda for Teflon
- Size
  - 10-15 Micron
- Quantity
  - 5-15 liter/min for fine work
  - 10-30 liter/min for usual cuts.
  - 50-100 liter/min for rough cuts

Typical AJM Parameters

- Medium
  - Dry air, CO_2, N_2
  - Quantity: 30 liter/min
  - Velocity: 150-300 m/min
  - Pressure: 200-1300 KPa
- Nozzle
  - Material: Tungsten carbide
  - Stand of distance: 2.54-75 mm
  - Diameter: 0.13-1.2 mm
  - Operating Angle: 60° to vertical

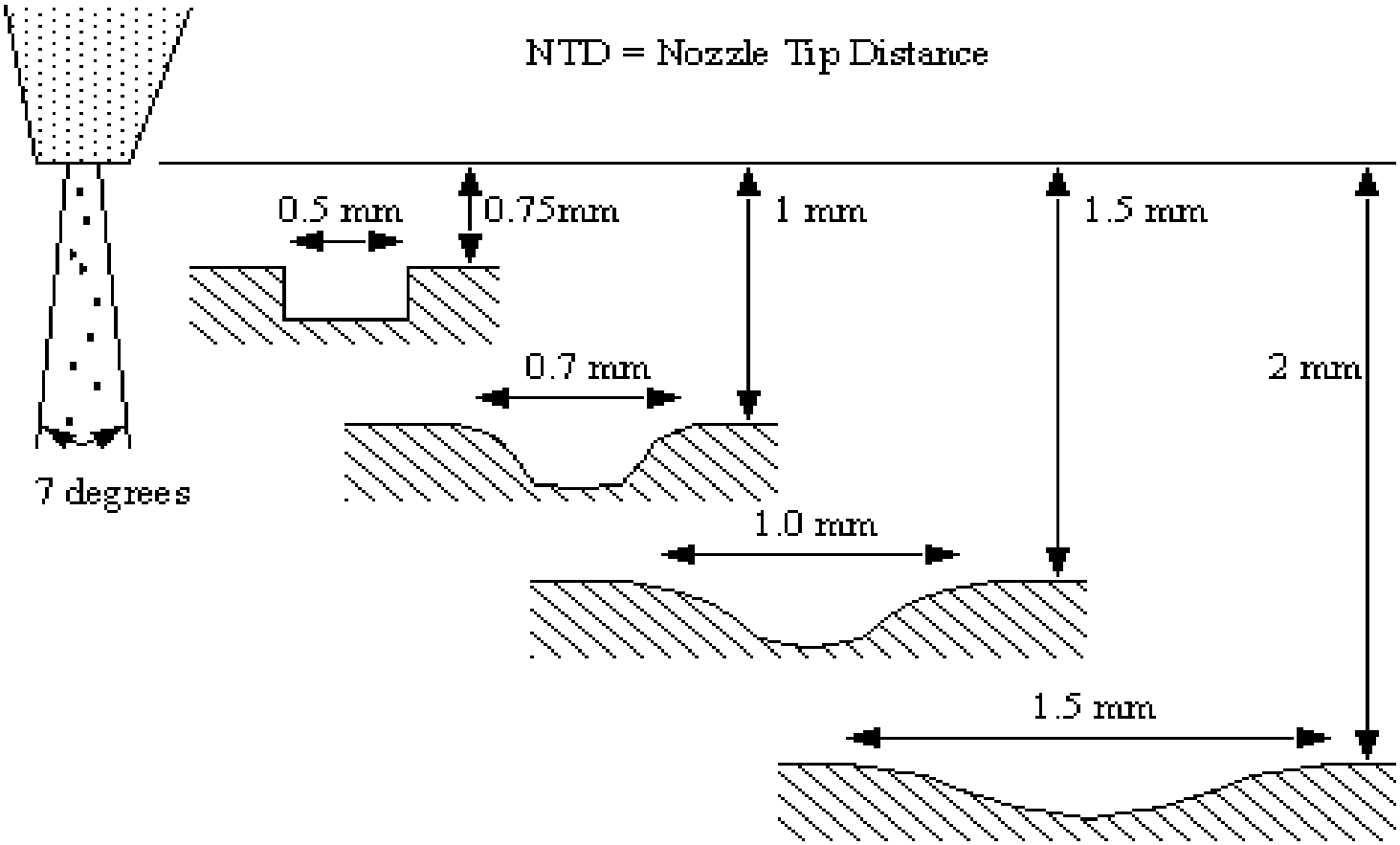
Typical AJM Parameters

- Factors affecting MRR:
  - Types of abrasive and abrasive grain size
  - Flow rate
  - Stand off distance
  - Nozzle Pressure

Nozzle

- The nozzle is one of the most vital elements controlling the process characteristics. Since it is continuously in contact with the abrasive grains flowing at a high speed, the material must be hard to avoid any significant wear.
- One of the most important factors in AJM is the distance between the work surface and the tip of the nozzle, normally called the nozzle distance.

Nozzle Tip Distance



Application

- For drilling holes of intricate shapes in hard and brittle materials
- For machining fragile, brittle and heat sensitive materials.
- AJM can be used for drilling, cutting, deburring, cleaning and etching.
- Micro-machining of brittle materials

Limitations

- MRR (Material removal rate) is rather low (around ~15 mm^3/min for machining glass).
- Abrasive particles tend to get embedded particularly if the work material is ductile.
- Tapering occurs due to flaring of the jet.

Advantages

- Extremely fast setup and programming.
- No start hole required.
- There is only one tool.
- Low capital cost.
- Less vibration.
- No heat generation in work piece.
- Environmentally Friendly.

Disadvantages

- Low metal removal rate.
- Abrasive powder cannot be reused.
- Tapper is also a problem.

Laser Beam Machining (LBM)

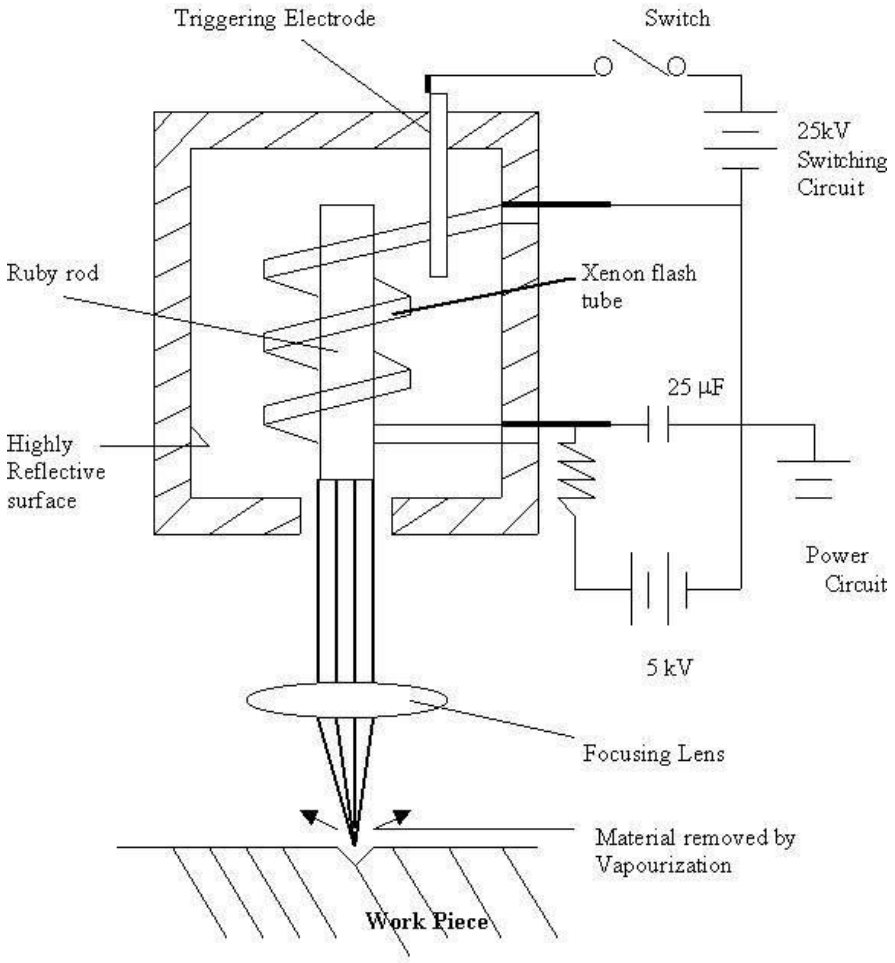
- Laser-beam machining is a thermal material-removal process that utilizes a high-energy, coherent light beam to melt and vaporize particles on the surface of metallic and non-metallic work pieces.
- Lasers can be used to cut, drill, weld and mark. LBM is particularly suitable for making accurately placed holes.

Laser Beam Machining (LBM)

- Conversion of electrical energy into heat energy to emit laser beam energy.
- Laser beam is focused on lance then create high energy the high energy concentration on workpiece then work piece is melt and vaporized of metal.

Working of Laser Beam Machining

- The diagram of LBM is shown in figure.
- Laser is stand for Light Amplification by Simulated Emulsion of Radiation.
- The workpiece is placed on the aluminum work table which material is hard not cut by laser beam.



Laser Beam Machining

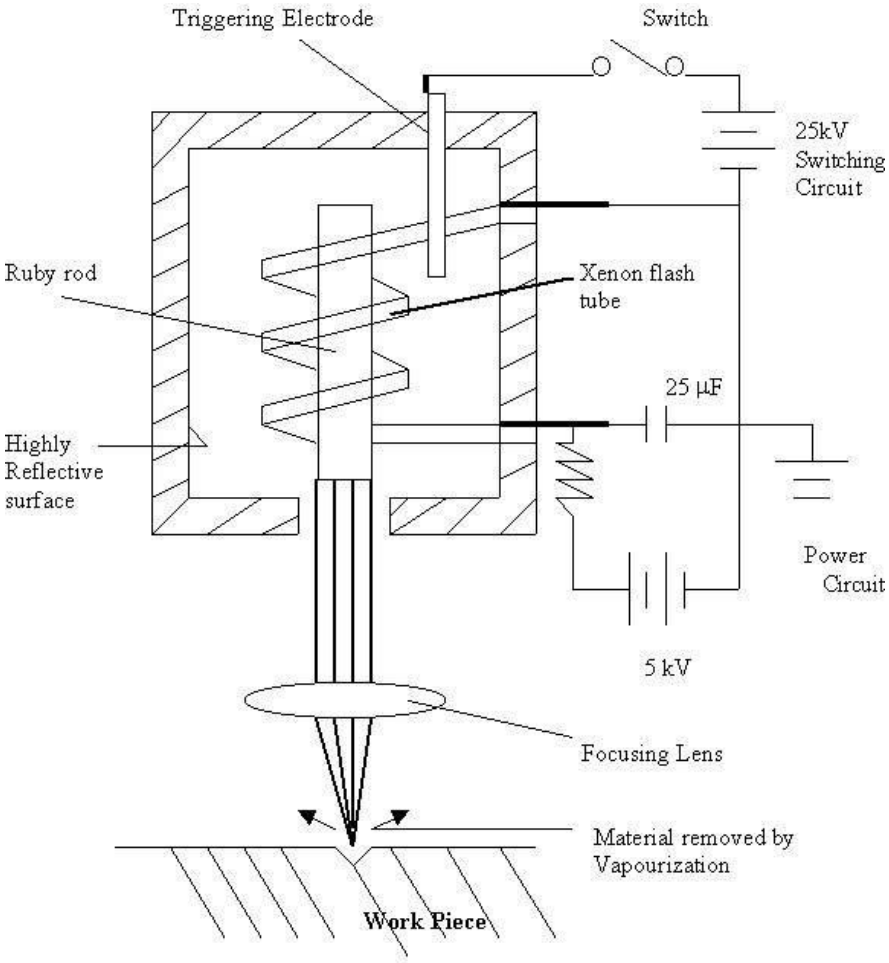
Working of Laser Beam Machining

1. A basic laser, like this red ruby laser, consists of a rod made of ruby crystals with a mirror on each end, and a flash tube.

2. A burst of light from the flash tube adds energy inside the rod, exciting the ruby atoms and producing light particles called photons.

Working of Laser Beam Machining

- Ruby rod is used into form of cylindrical crystal both ends of ruby rod are finished to optical tolerance.
- The flash lamp wound around the ruby rod and connected to power supply.



Laser Beam Machining

Working of Laser Beam Machining

- The ruby rod becomes high efficient on low temperature and low efficient on high temperature. It is thus continuous cooled with water, air or liquid nitrogen.
- When the light beam has been amplified sufficiently and intensity beam of light comes out form partially reflected end it is focused on the workpiece at the focused very high temperature which vaporized and removes the metal on workpiece.

Laser Beam Machining (LBM)

- Mechanics of material removal : Melting, Vaporization
- Medium : Normal atmosphere
- Tool : Higher power laser beam
- Maximum material removal rate: 5 mm^3/min
- Specific power consumption : 1000 W/mm^3/min
- Materials application : All materials
- Shape application : Drilling fine holes
- Limitations : Very high power consumption, cannot cut materials with high heat conductivity and high reflectivity

Applications

- LBM can make very accurate holes as small as 0.005 mm in refractory metals ceramics, and composite material without warping the work pieces.
- It is used for welding of thin metal sheet.
- Laser can be used for cutting as well as drilling.
- Heat treatment.
- It is used for cutting complex profile.

Limitations

- Uneconomic on high volumes compared to stamping
- Limitations on thickness due to taper
- High capital cost
- High maintenance cost
- Assist or cover gas required

Advantages

- Very hard and abrasive material can be cut.
- Sticky materials are also can be cut by this process.
- It is a cost effective and flexible process.
- High accuracy parts can be machined.
- No cutting lubricants required
- No tool wear
- Narrow heat effected zone
- No contact between tool and workpiece.

Disadvantages

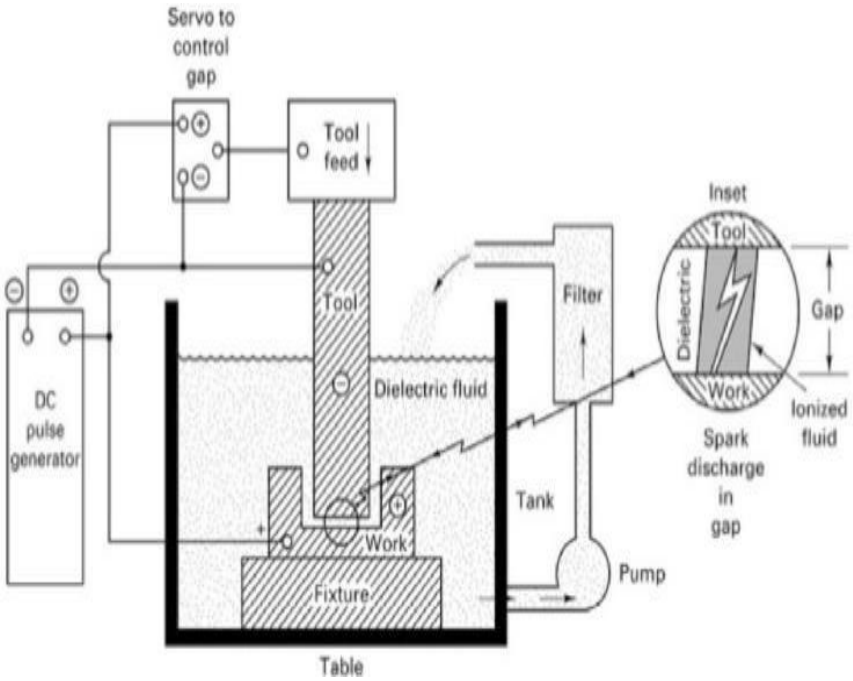
- Investment cost is more.
- Skilled operator is required.
- Operating cost is more.
- Flash lamp life is too short.

Electrical Discharge Machining (EDM)

- Electrical discharge machining (EDM), sometimes also referred to as spark machining, spark eroding, burning, die sinking, wire burning or wire erosion, is a manufacturing process whereby a desired shape is obtained using electrical discharges (sparks).
- Material is removed from the workpiece by a series of rapidly recurring current discharges between two electrodes, separated by a die-electric liquid and subject to an electric voltage. One of the electrodes is called the tool-electrode, or simply the "tool" or "electrode", while the other is called the workpiece-electrode, or "workpiece".

Working of EDM

- EDM is thermal erosion process whereby material is melted and vaporized from an electrically conductive workpiece immerse in a liquid dielectric with a series of spark discharge between the tool electrode and the workpiece created by a power supply.

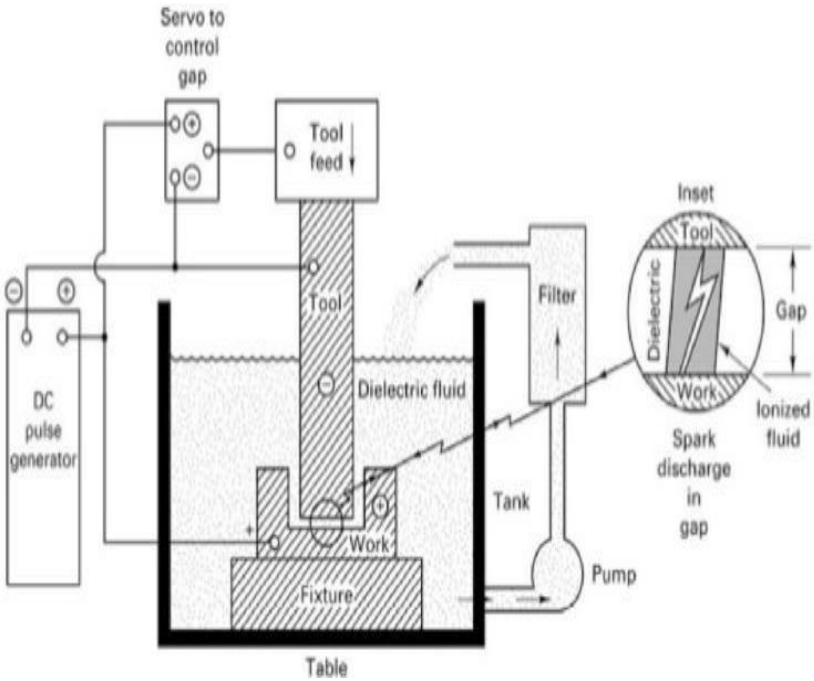


Working of EDM

Dielectric fluid in EDM machines creates an insulating gap to control electrical discharge, acts as a coolant to prevent thermal damage, and flushes away microscopic eroded particles.

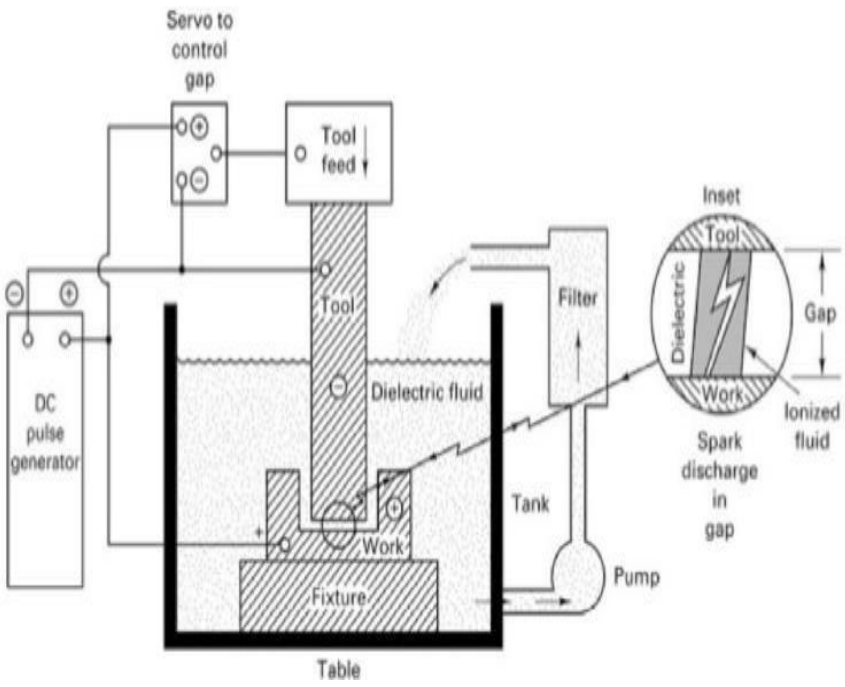
Working of EDM

- The electrode and the workpiece are separated by a dielectric medium.
- The dielectric medium is like as kerosene, paraffin or light oil.
- The strong electrostatic field between the electrode and workpiece produce emission of electrons from the cathode.



Working of EDM

- In this gap between tool and workpiece get ionized. The liquid is force to sparking zone.
- Due to high temperature, the metal at the sparking zone melts instantaneously.
- The material of the tool is usually a material which conduct electricity and which can be easily shaped.



Advantages

- Smaller holes can be easy machined.
- No contact between tool and workpiece then tool life is increase.
- Any complex shape can be machined.

Disadvantages

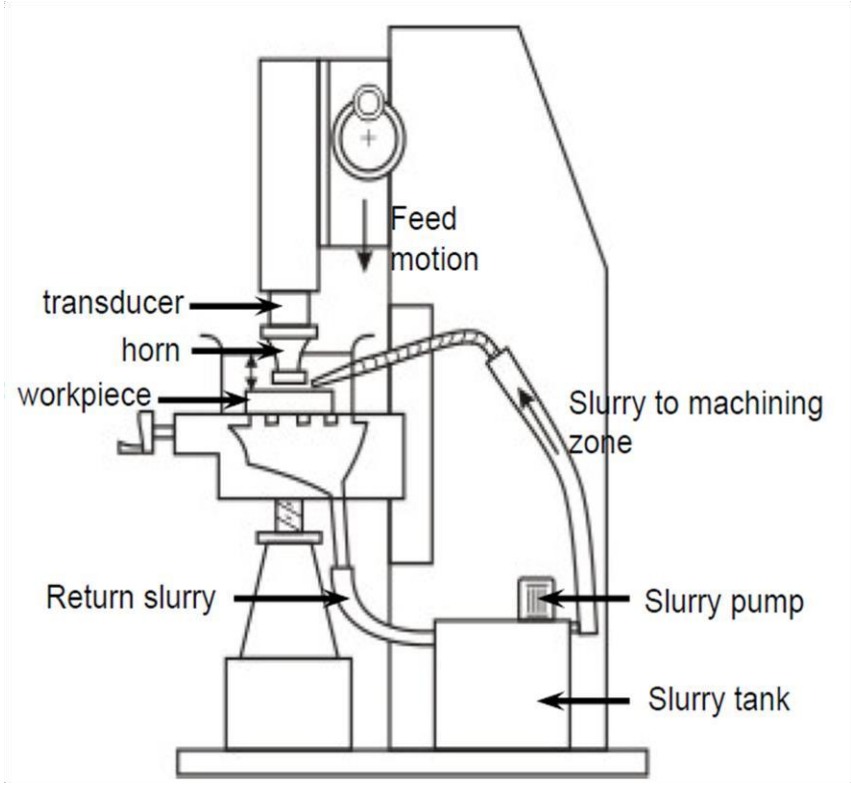
- Tool life is not longer.
- Power consumption is high.
- Cycle time is more

Ultrasonic machining (USM)

- In this method with the help of piezoelectric transducer tool is vibrate at high frequency in a direction normal to the surface being machined abrasive slurry are used for the remove the metal from workpiece.

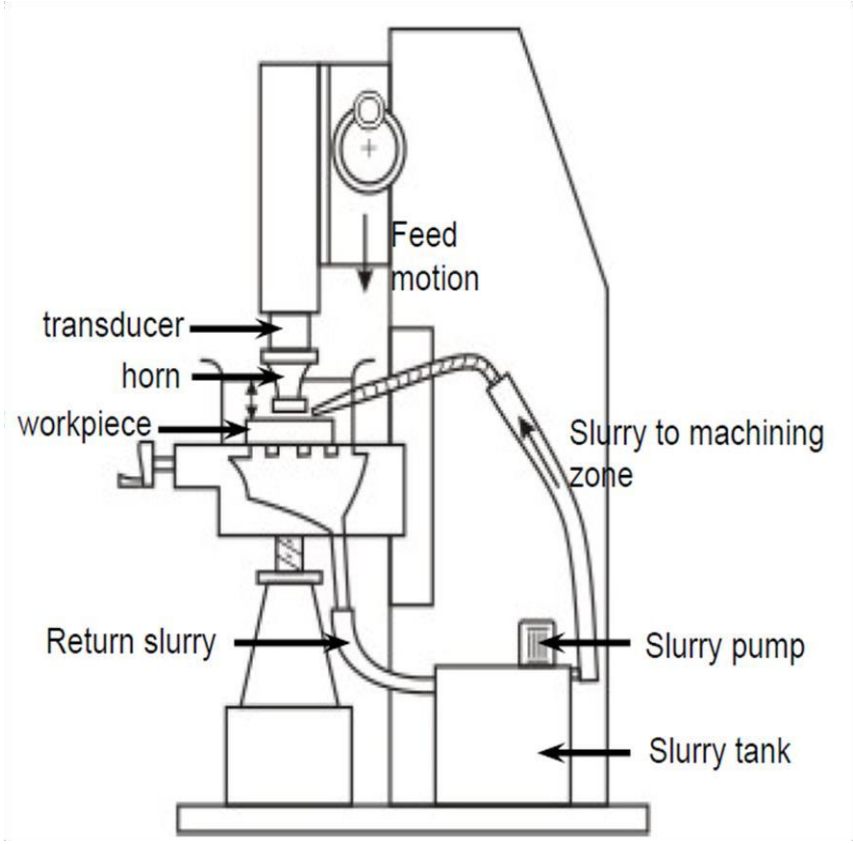
Working of Ultrasonic machining

- The USM diagram shown in figure.
- In ultrasonic machining a tool vibrate longitudinally at 20 to 30 kHz with amplitude between 0.01 to 0.06 mm is pressed on to the work surface with light force.
- The electronic oscillator and amplifier is also known as generator.



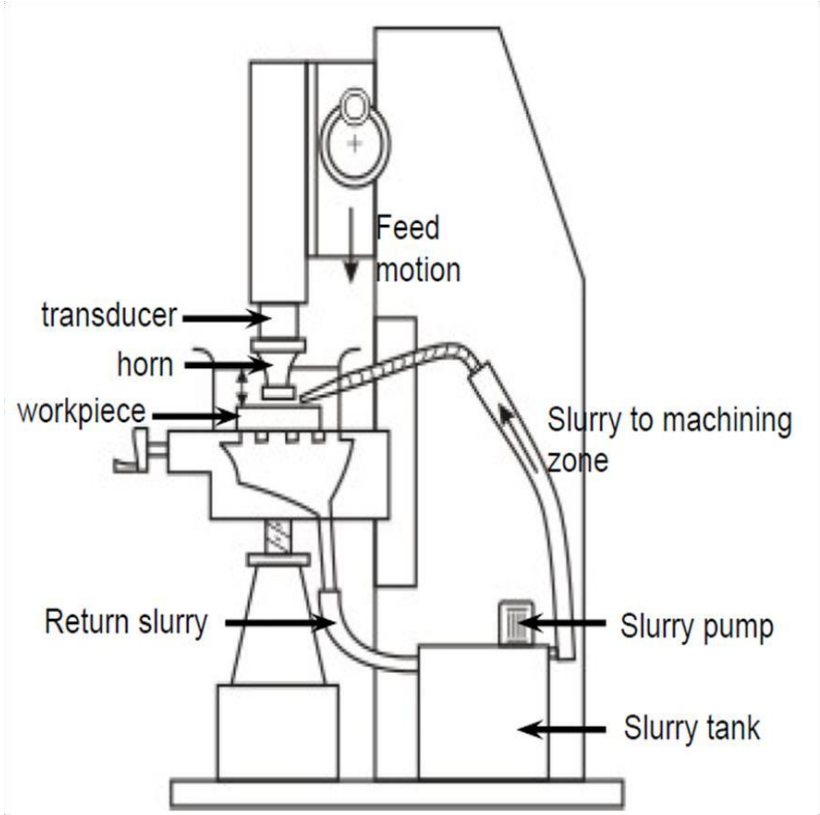
Working of Ultrasonic machining

- It converts the electrical energy of low frequency to high frequency.
- At the time high frequency current is passed through the coil therefore change in electromagnetic field which produces longitudinal strain.



Working of Ultrasonic machining

- As the tool vibrate with specific frequency the abrasive slurry mix with water and grain of definite proportion is made to flow under pressure through the tool workpiece interface. The flow of slurry through the work tool interface actually causes thousand of microscopic grain to remove the work material by abrasion.



Ultrasonic machining (USM)

- Mechanics of material removal : Brittle fracture caused by impact of abrasive grains due to tool vibrating at high frequency.
- Medium : Slurry
- Tool : Soft Steel
- Gap : 25-40 µm
- Frequency : 15-30kHz
- Amplitude : 25-100 µm
- Materials application : Metals and alloys, semiconductors, non-metals
- Shape application : Round and irregular holes
- Limitations : Very low MRR, tool wear, depth of holes and cavities small

Applications

- USM is best suitable for hard, brittle material, such as ceramics, carbides, glass, precious stone etc.

Thank You